Chapter 4 — Biomolecules

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. High specific heat of water helps: (a) Stable temperature (b) Cell walls (c) DNA storage (d) Evaporation
2. Cohesion of water is due to: (a) Hydrogen bonding (b) Peptide bonding (c) Ionic bonding (d) Covalent bonding
3. Glucose and fructose are: (a) Polymers (b) Ions (c) Isomers (d) Isotopes
4. Storage carbohydrate in animals is: (a) Chitin (b) Glycogen (c) Starch (d) Cellulose
5. Bond between amino acids is: (a) Phosphodiester bond (b) Hydrogen bond (c) Peptide bond (d) Glycosidic bond
6. Phospholipids are key components of: (a) Ribosomes (b) Chromosomes (c) Cell walls (d) Cell membranes
7. In DNA, adenine pairs with: (a) Guanine (b) Uracil (c) Thymine (d) Cytosine
8. Lipoproteins help in: (a) DNA replication (b) Oxygen transport (c) Lipid transport in blood (d) Cell wall synthesis

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is water an excellent solvent for biological systems?
2. How does the high heat of vaporization of water help organisms?
3. Differentiate between starch and glycogen.
4. How does the primary structure of a protein determine its function?
5. Why are phospholipids able to form bilayers in water?
6. Differentiate between DNA and RNA.

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the properties of water that make it essential for life. **(4)**
(b) Classify carbohydrates and describe their biological importance. **(4)**
Q.4 (a) Describe the structural levels of proteins. **(4)**
(b) Explain the double helical structure of DNA. **(4)**

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